



Indywidualny Plan Badawczy (IPB) Individual Research Plan (IRP)

1. Podstawowe dane / Basic data

Podstawowe informacje dotyczące doktoranta i realizowanej rozprawy doktorskiej. W przypadku promotora spoza Politechniki Wrocławskiej należy podać afiliację. / *Basic information on the doctoral student and the doctoral dissertation underway. In the case of a supervisor from outside Wrocław University of Science and Technology, affiliation should be provided.*

Imię i nazwisko / Name, surname	Grzegorz Piotrowski
Dyscyplina kształcenia / Discipline of education	informatyka techniczna i telekomunikacja / information and communication technology
Wydział / Faculty	W4 Wydział Informatyki i Telekomunikacji / Faculty of Information and Communication Technology
Katedra / Department	K46 Katedra Sztucznej Inteligencji / Department of Artificial Intelligence
Data rozpoczęcia kształcenia / Date of commencement of the doctoral education	01.10.2025
Nr ORCID	0009-0004-6013-0461
Promotor/ promotorzy / Supervisor(s)	dr hab. inż. Tomasz Jan Kajdanowicz, prof. uczelni, Politechnika Wrocławska (K46, W4), ORCID 0000-0002-8417-1012
Promotor pomocniczy / Assistant supervisor	—

2. Temat rozprawy doktorskiej / Topic of doctoral dissertation

Propozycja tematu rozprawy doktorskiej. Przedstawiony temat nie musi stanowić ostatecznego tytułu, jaki zostanie nadany rozprawie doktorskiej złożonej na zakończenie kształcenia w Szkole Doktorskiej. / *Proposed topic for a doctoral dissertation. The presented topic does not have to constitute the final title of the doctoral dissertation submitted at the end of education at the Doctoral School.*

Metody uczenia reprezentacji dla generalizacji polityk uczonych w modelach świata do rzeczywistości w fizycznej sztucznej inteligencji / Representation learning methods for the simulation-to-reality generalization of policies trained in world models for physical AI

3. Harmonogram przygotowania rozprawy doktorskiej / Schedule for the preparation of the doctoral dissertation

Harmonogram przygotowania rozprawy doktorskiej z wyszczególnieniem etapów i miejsc prowadzonych badań, w tym terminów zakończenia badań cząstkowych (semestr kształcenia) i opracowania ich wyników oraz opcjonalnie w zależności od rodzaju prowadzonych badań, wybranych między innymi spośród następujących:

- realizacja badań (w tym przygotowanie stanowiska badawczego, przygotowanie materiałów do badań, wykonanie pomiarów/eksperymentów, opracowanie otrzymanych wyników),
- przygotowanie i wysłanie artykułu naukowego do redakcji czasopisma naukowego,



- wysłanie zgłoszenia patentowego do właściwego urzędu,
- opracowanie propozycji wdrożenia dla przemysłu,
- udział w konferencjach, warsztatach lub szkołach letnich/zimowych i prezentacja wyników badań,
- przygotowanie wniosku i wystąpienie o grant,
- mobilność krajowa lub zagraniczna doktoranta (w tym konsultacje naukowe i wizyty studyjne w ośrodkach krajowych lub zagranicznych).

A schedule for the preparation of the doctoral dissertation detailing the stages and locations of the research to be conducted, including the dates for completing the sub-studies and developing their results, and optionally, depending on the type of research to be conducted, selected among the following:

- *execution of the research (including preparation of the research site, preparation of materials for the research, execution of measurements/experiments, elaboration of the obtained results),*
- *preparation and submission of a scientific article to the editor of a scientific journal,*
- *submission of a patent application to the relevant office,*
- *development of an implementation proposal for the industry,*
- *participation in conferences, workshops or summer/winter schools and presentation of research results,*
- *preparation of an application and application for a scholarship,*
- *domestic or foreign mobility of the doctoral student (including scientific consultations and study visits to domestic or foreign centres).*

Okres realizacji zadania (semestr kształcenia) / Task implementation period (semester)	Zwięzły opis zadania / Brief description of the task
Semestr 1	Doctoral School courses. Reading the literature and considering possible research directions
Semestr 2	Doctoral School courses and literature review. Choosing the dissertation topic with the supervisor and preparing the Individual Research Plan
Semestr 3	Setting up the research environment with world models, a physics simulator and public robot datasets. Initial experiments on harmful errors of world-model simulations (RQ1)
Semestr 4	Further experiments on RQ1. Preparing and submitting the first publication to a 200-point venue in machine learning or robot learning, such as ICLR, RSS or IEEE RA-L. First experiments on RQ2
Semestr 5	Experiments on localizing the gap inside policies and aligning representations (RQ2). Preparing a scientific publication
Semestr 6	Experiments on selecting real rollouts (RQ3) and on transfer to unseen scenes and between tasks (RQ4). Preparing a publication and, if access allows, tests on a real robot
Semestr 7	Final experiments and submitting a paper to a conference or journal. First draft of the dissertation
Semestr 8	Completing and revising the doctoral dissertation, then submitting it

4. Termin złożenia rozprawy doktorskiej / Date of the doctoral dissertation submission



Planowany termin złożenia rozprawy doktorskiej nie może przekraczać daty zakończenia ósmego semestru kształcenia w Szkole Doktorskiej. / A planned date for submission of the dissertation, which must not exceed the date of completion of the eighth semester of education in the Doctoral School.

September 2029

(rekomendowana data to ostatni dzień z okresu 4 lata kształcenia w Szkole Doktorskiej, np. jeżeli kształcenie rozpoczęło się 1.10.2025, to proszę wpisać 30.09.2029 /

The recommended date, which marks the final day of the four-year doctoral training period. For example, if your training began on October 1st, 2025, please enter September 30th, 2029)

5. Uzasadnienie wyboru tematu rozprawy doktorskiej / Justification for choosing the topic of the doctoral dissertation

Opis zawierający uzasadnienie wyboru tematu rozprawy doktorskiej uwzględniający potencjalne obszary zastosowania planowanych wyników rozprawy doktorskiej (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description containing the justification for the selection of the topic of the doctoral dissertation, taking into account the potential areas of application of the planned results of the doctoral dissertation (at most 1 page, font 11, line spacing 1).

Robots that learn by trial and error need far more attempts than a real robot can make safely or cheaply, so they are usually trained in simulation. Hand-built physics simulators are fast, but building a realistic copy of each new place takes a lot of work, and their images still look different from real camera images. A newer option is a learned world model [1]. It is a neural network that predicts what the robot will see after it takes an action. Recent world models are trained on large amounts of real video and can serve as simulators. UniSim [2] trained robot policies only inside such a model and then used them on real robots. Navigation World Models [3] imagine how a walk through an unfamiliar place would look from a single image of it, and NVIDIA's Cosmos models [4] are released as a base for building such simulators.

The weak point of this approach is that an imagined rollout is never exact. The model may ignore an action and show the robot standing still. It may let a gripper pass through an object or show an object being lifted without proper contact. Small errors also add up over long rollouts, until the scene drifts away from anything physically possible [5],[6]. A policy that succeeds in the world model can therefore fail on a real robot. In the first study that moved a policy of this kind from purely synthetic training to a real robot arm, it succeeded in about a third of the trials [7]. It is still not known which of these errors actually harm a policy, and why. Better-looking video does not answer this, because the visual quality of a world model does not reliably predict whether policies succeed in it [6].

I want to answer this question by looking inside the models. A neural network does not act on the image directly. It first turns it into internal features, called representations, and decides from them. LeCun argued that a world model should also predict in such a representation space rather than in pixels, so that it can ignore details that cannot be predicted and do not matter for control [8]. The V-JEPA 2 model [9] showed that a world model of this kind can plan the movements of a real robot arm. My working assumption follows the same idea. An error of the world model that leaves the representations unchanged should do little harm, while an error that changes the features the decision depends on should cause failures [10]. This gives a measurable way to study the gap. First, I will check which errors of a world-model simulation change the representations in a harmful way. Then I will find the stage inside the policy at which real inputs lose information needed for the task, and correct the representations there. I will also study how to choose the few real rollouts that best correct both the world model and the policy, and whether the gains hold in new scenes and in a second task. I will work on two tasks of equal weight, robot navigation and robotic manipulation.

The answer would show which properties of a world model must be accurate for a given task and where extra real data are worth collecting. In practice, a robot could learn a new job in a warehouse, a hospital or a home mostly from predicted experience and then work there reliably after little real-



world adaptation. The same questions arise in autonomous driving, where generative world models already produce scenarios for training and testing, and in models that predict the future and choose actions within one network.

6. Zarys aktualnego stanu badań w tematyce rozprawy doktorskiej / **Outline of the current state of research in the topic of the doctoral dissertation**

Przegląd literaturowy zawierający opis aktualnego stanu badań w tematyce rozprawy doktorskiej uwzględniający odwołania do najważniejszych pozycji literaturowych związanych z tematyką rozprawy doktorskiej (maks. 2 strony, czcionka 11 pt., interlinia 1). / A literature review containing a description of the current state of research in the subject of the doctoral dissertation, including references to the most important literature items related to the subject of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

The idea of learning a model of the environment and training a controller inside its predictions goes back at least to Ha and Schmidhuber [1], whose agent learned to play a game inside its own "dream" and then played the real game. DreamerV3 [11] turned training in imagination into a general method that works with one fixed configuration across more than 150 tasks. Such models are small and learned for one environment. Since 2023, large generative models trained on internet and robot video have been used as interactive simulators of the real world. UniSim [2] learned a simulator from mixed internet, robot and navigation data, and policies trained purely in it transferred to the real world without further training. Navigation World Models (NWM) [3] predict the egocentric video of a moving robot and plan by simulating candidate trajectories. NVIDIA's Cosmos platform [4] provides open world foundation models that are meant to be post-trained into world models for particular robots and vehicles.

NVIDIA also combines these models with its physics simulators, Isaac Sim and Isaac Lab [12]. Cosmos Transfer, part of the Cosmos family, turns simulator renderings such as depth or segmentation into photorealistic video with the same geometry and motion, so the simulated actions remain valid. DreamGen [13] fine-tunes a video world model on the target robot, generates videos of new behaviours, labels them with pseudo-actions and trains a policy on them. With these data, NVIDIA's GROOT N1 policy learned 22 new behaviours from teleoperation data of a single task, and its success improved on three real robots.

A newer line merges the world model and the policy. A World Action Model (WAM) is a single network that predicts both future states of the world and the robot's actions. The term was set by DreamZero [14], which reported that such a model generalized to new tasks and environments more than twice as well as vision-language-action models (VLAs). VLAs map images and instructions directly to actions. The sim-to-real transfer of WAMs has barely been studied. The first work to report it [7] fine-tuned a Cosmos video model into such a policy, trained it only on synthetic demonstrations and reached 35% average success on a real Franka arm without any real training data.

LeCun's position paper [8] argues that a world model should not predict pixels. In his Joint Embedding Predictive Architecture (JEPA), the model encodes the current and the future observation and predicts the future representation, so it can drop unpredictable detail such as texture or sensor noise. DINO-WM [15] predicts frozen pretrained DINOv2 features and plans without reconstructing pixels. V-JEPA 2 [9] was pretrained on more than a million hours of video. Its action-conditioned version, V-JEPA 2-AC, was post-trained on less than 62 hours of robot video from the DROID dataset and planned pick-and-place on real Franka arms in labs it had not seen, without rewards. SkyJEPA [16] trained a JEPA world model on simulated data and controlled a real quadrotor without real-world training. These works show that latent world models can transfer to reality, but they do not measure the gap between simulated and real inputs in the latent space or relate it to the success of a policy.

World models are also used to evaluate policies instead of running them on a robot. Ctrl-World [17], trained on DROID, ranked policies without real rollouts and improved a policy by fine-tuning it on



imagined successful trajectories. A world simulator based on Veo [18] predicted the relative performance of Gemini Robotics policies, which was checked with more than 1600 real trials. World-in-World [6] tested world models in closed loop on navigation and manipulation and found that visual quality does not guarantee task success, while how well a model follows actions matters more. VLAW [5] found that current world models lack the physical accuracy needed to improve policies, because they are trained on demonstrations without failures and miss small contact details. For physics simulators, the SIMPLER study [19] showed that scenes visually matched to real images can rank policies in nearly the same order as real runs. These studies measure the errors of a world model at its output or through the final correlation with real success. They do not show which type of error changes the policy, or how.

The gap between a world model and reality is a case of distribution shift. Domain adaptation theory bounds the error on the target domain by the error on the source domain, a discrepancy between the two distributions and the joint error of the best single model on both domains [20]. I treat this bound only as motivation. Making features invariant to the domain does not target the joint error and can even increase it [21]. Domain randomization varies simulator parameters such as textures so that reality looks like one more variation [22]. It changes global settings of a hand-built simulator, while the errors of a world model depend on the action, the contact and the length of the rollout. Co-training on simulated data together with a small real set improved real-world manipulation [23]. A mechanistic analysis of such co-training showed that it aligns the two domains but keeps them distinguishable [24], so full alignment everywhere is not the goal.

It is also unclear where inside a policy the gap arises. Probing, which reads information out of hidden states with simple classifiers, showed that fine-tuning for actions degrades the visual representations of VLAs [10]. Surgical fine-tuning showed that the best layers to adapt depend on the type of shift [25]. I found no work that locates, in policies trained in world models, the stage at which task information available for imagined inputs is lost for real inputs.

When a small amount of real data is collected, it is used in different ways. VLAW [5] fine-tunes the world model on real rollouts and then improves the policy with synthetic data from it, without choosing which rollouts to collect. TwinRL [26] uses a digital twin to find failure-prone configurations for real rollouts that improve the policy. How to choose real rollouts under a fixed budget so that they are most useful for the world model and the policy is still open.

Finally, the generalization gap of visual manipulation policies can be split into factors such as lighting or camera placement, and the ordering of these factors by difficulty was largely the same in simulation and on a real robot [27]. It is not known whether a gap measured in latent space before a method is applied can predict whether its gain carries over to a new scene or task, or whether it does better than simple indicators such as the video quality of the world model. My research questions in §7 address these open points. RQ1 asks which errors harm the policy, RQ2 where inside the policy the harm appears, RQ3 which real rollouts to collect and RQ4 whether the gains and the latent measures transfer.

(Please see section 12 for citations used in this document)

7. Pytania i hipotezy badawcze / Research questions and hypotheses

Opis najważniejszych pytań i hipotez badawczych związanych z realizowaną rozprawą doktorską (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description of the most important research questions and hypotheses related to the doctoral dissertation (at most 1 page, font 11, line spacing 1).

The aim of my dissertation is to develop representation learning methods that improve the real-world generalization of robot policies trained in simulations created by learned world models. The policies should work reliably on real inputs and in scenes not seen during training, and the methods should work with unchanged settings for both navigation and manipulation.



I will not have my own robot at the start and will work on academic GPU clusters, so I will use existing pretrained world models and policies and fine-tune them rather than train large models from scratch. As the reference for reality I will use recorded real robot data and published real-robot results, and a real robot if access allows. This reference is a proxy, and I will report which part of the gap it measures. I will study robot navigation and robotic manipulation and measure all results on held-out scenes that were not used for training or model selection.

My research questions (RQ) and hypotheses (H) are as follows.

1. RQ1. Which errors of a world-model simulation, such as weak action following, wrong contact physics, appearance errors and long-horizon drift, varying across situations, harm the generalization of policies trained in it?
2. H1. The harm of an error is the drop in success on held-out real data of a policy trained in a world model with this error, compared with the same policy trained without it. This harm is predicted better by how much the error changes the latent representation of a fixed pretrained encoder or of a JEPA-style world model than by pixel-level video quality. Also, targeted augmentation of the harmful error types generalizes better than uniform augmentation of equal strength.
3. RQ2. At which stage of a policy trained in a world model do real inputs first lose task information that is available for imagined inputs? I will look for it with linear probes checked against control tasks and test with causal interventions whether the policy uses this information.
4. H2. Aligning imagined and real representations at that stage reduces the gap more than the same alignment at the input, at the final features or at all stages, given the same real data and training effort.
5. RQ3. Which few real rollouts, chosen under a fixed budget with the signals of RQ1 and RQ2, best correct both the world model and the policy? I will compare my selection with random and failure-driven selection and with the same amount of unselected real data used as in VLAW [5].
6. RQ4. Do the gains transfer with unchanged settings to unseen scenes and between navigation and manipulation? Can a gap measured in latent space, split into an observation gap in the encoder and a dynamics gap in the predictor as the JEPA view suggests, predict this transfer better than simple indicators such as the rank correlation of world-model success with real success or video quality?
7. Fallback. If pixel-level quality predicts harm better than the change in representations, I will guide augmentation and data selection with the uncertainty of the world model and the sensitivity of the policy's actions instead. This fallback is weaker, because a world model can be confidently wrong, for example about contact after actions far from its training data.

8. Wkład spodziewanych wyników dla rozwoju dyscypliny naukowej / Contribution of the expected results to development of scientific discipline

Krótki opis spodziewanego wkładu wyników rozprawy doktorskiej dla rozwoju dyscypliny naukowej, w której realizowana jest rozprawa doktorska (maks. 1 strona, czcionka 11 pt., interlinia 1). / Brief description of the expected contribution of the results of the doctoral dissertation to the development of the scientific discipline in which the doctoral dissertation is carried out (at most 1 page, font 11, line spacing 1).

The main expected contribution of my dissertation is a set of representation learning methods that help robot policies trained in world-model simulations generalize to reality and to unseen scenes. Together with the methods, I want to explain how the errors of a world model reach a policy. I want to learn which types of error matter, at which stage of the policy they turn into lost task information and how much well-chosen real data is needed to correct them.

This is timely because world models are becoming a common way to create training data and to evaluate policies, and World Action Models now put the world model and the policy into one network. My results should show which properties of a world model must be accurate for a given



task. This is useful both to those who build world models and to those who train policies in them. Because the questions concern learning under distribution shift, the results on where to align representations and which data to select may also be useful in domain adaptation and active learning. I plan to release the code, the evaluation protocol and, where licences allow, the trained models, so that others can compare methods in the same way.

9. Planowane metody badawcze / Planned research methods

Krótki opis metod badawczych, które są będą użyte w czasie realizacji rozprawy doktorskiej dla (maks. 2 strony, czcionka 11 pt., interlinia 1). / Brief description of the research methods that will be used during the implementation of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

Most of my work will consist of training and fine-tuning policies inside learned world models, measuring how they perform on real data and designing methods that analyse and align the representations they learn. No established theory covers the generalization of policies trained in world models, so the work will be mostly empirical. Domain adaptation theory [20] will be my motivation, and where possible I will check whether an alignment lowers the discrepancy between imagined and real features without raising the joint error.

I will use PyTorch for training, Git with GitHub repositories for version control and Hugging Face for downloading and publishing models and data. For navigation I will start from NWM [3]. For manipulation I will use DINO-WM [15], V-JEPA 2-AC [9] or the small LeWorldModel as latent world models, and Ctrl-World [17] or Cosmos-Predict [4] as video world models. I will use open weights where they exist and respect their licences, some of which allow only non-commercial use. As a physics simulator for comparison I will use NVIDIA Isaac Lab 3.0 in its kit-less mode with the Newton physics engine, which runs on H100 GPUs. I will use Cosmos Transfer to add photorealistic appearance to simulator renderings. Isaac Sim does not support A100 or H100 GPUs, so I will use it only on an RTX workstation if one is available.

Since I will not have my own robot at the start, I will evaluate mainly on public data. For manipulation I will use DROID, on which Ctrl-World and V-JEPA 2-AC were trained, and BridgeData V2, whose WidowX setup is one of the robot setups modelled in SIMPLER [19]. For navigation I will use RECON and SCAND, two of the real robot datasets on which NWM was trained. Held-out trajectories and scenes of these datasets give real observations paired with imagined ones that start from the same state and follow the same actions, which I need for probing and alignment. I will also check whether the ranking of policies in a world model agrees with published real-robot results, from SIMPLER and from the real evaluations reported for the world models. Evaluation inside a world model or on recorded data is only a proxy for reality. It leaves out parts of the gap, such as the robot's own actuation and camera, and I will state this with every result. If access allows, I will use a real robot in a university laboratory to check selected results.

For RQ1 I will inject one type of error at a time into a world model, or remove it by replacing imagined frames with real or physics-simulated ones. The error types are ignored or delayed actions, wrong contact outcomes, appearance changes and drift over long rollouts. I will train or fine-tune a policy in each version and measure the change in its success. I will compare the change in the latent representation of a fixed pretrained encoder, such as DINOv2 or V-JEPA 2, with pixel-level video metrics such as PSNR, LPIPS and FVD by their rank correlation with harm. I will also test targeted augmentation against uniform augmentation of equal strength (H1). For RQ2 I will train linear probes, that is, linear models that read task information such as the direction to the goal or the position of an object, on the hidden states of each stage of the policy for imagined inputs, and test them on the paired real inputs. Control tasks with random labels will show that the probes do not simply memorize. To test whether the policy uses the lost information, I will replace the component of the real hidden state along the probe directions with the one from the paired imagined input and check whether the action improves. I will compare this with replacing the same number of random directions. I will then add an alignment loss, such as maximum mean discrepancy or a contrastive loss



between paired features, at the identified stage. I will compare it with the same loss at the input, at the final features, at all stages and at layers chosen as in surgical fine-tuning [25] (H2). For RQ3 I will compare rules for selecting real rollouts with random selection, failure-driven selection as in TwinRL [26] and unselected data as in VLAW [5], at equal budgets. I will also compare correcting both the world model and the policy with correcting only one of them. For RQ4 I will apply the methods with unchanged settings to held-out scenes and to the other task. I will measure the latent gap as an observation term, the distance between the embeddings of matched imagined and real states, and a dynamics term, the error of the world model's latent prediction on real transitions. I will test whether these terms predict the gains better than the rank correlation of world-model success with real success or video quality, using Spearman rank correlation across scenes with permutation tests.

For every policy I will measure the success rate in the world model and in the evaluation setting of each task, the gap between imagined and real performance and the rank correlation of world-model results with real results. I will compare my methods with domain randomization [22], co-training on imagined and real data [23] and the baselines named above, with the same number of training steps and the same amount of real data for every method. To get reliable results I will repeat runs over random seeds and scenes, use statistical tests for comparisons and run ablation studies. Data for training, model selection and final evaluation will be kept separate.

I will run experiments on the H100 nodes of the Wrocław Centre for Networking and Supercomputing (WCSS), with jobs managed by SLURM, and on PLGrid resources. I intend to use AI coding assistants like Claude Code to speed up writing and testing code. They will also help with routine work such as monitoring long experiments and proofreading.

I plan to publish my results at conferences and in journals worth 200 points on the ministerial list. In machine learning these are NeurIPS, ICML and ICLR, in computer vision CVPR, ICCV and ECCV, and in robot learning RSS and the journal IEEE RA-L. Where possible, I will also post the papers as arXiv preprints.

10. Streszczenie popularnonaukowe / Abstract for general public

Streszczenie popularnonaukowe należy przygotować w dwóch językach: polskim i angielskim. Wersje językowe muszą być tożsame. Opis powinien być napisany językiem popularnonaukowym i zawierać cel rozprawy, opis planowanych badań naukowych oraz powodów, dla których podjęta została ta tematyka badawcza, a także najważniejsze spodziewane efekty (maks. 1 strona, czcionka 11 pt., interlinia 1). / *The description for general public must be in Polish and in English. The language versions must be identical. The description should be written for general public and should include the doctoral dissertation goal, description of research, reasons for attempting a particular research topic and substantial results expected (at most 1 page, font 11, line spacing 1).*

Streszczenie popularnonaukowe

Roboty coraz częściej uczą się we „śnie”, czyli w kopii świata przewidywanej przez sieć neuronową zwaną modelem świata, ponieważ nauka w prawdziwym świecie jest powolna, kosztowna i czasem niebezpieczna. Sieć uczy się z nagrań wideo, co zwykle dzieje się po ruchu robota, ale jej sen nigdy nie jest idealny. Może zignorować ruch, pozwolić, by chwytak przeszedł przez przedmiot, albo oddalić się od rzeczywistości. Dlatego robot, który dobrze radzi sobie we śnie, w prawdziwym świecie często zawodzi, zwłaszcza w nowych miejscach. Chcę ustalić, które błędy snu szkodzą robotowi i gdzie wewnątrz jego sieci neuronowej zamieniają się w błędne działania. Chcę też sprawdzić, które nieliczne prawdziwe próby najlepiej poprawiają zarówno sen, jak i robota. Zbadam nawigację robotów oraz chwytanie i przedstawianie przedmiotów. Efektem mają być metody, dzięki którym roboty będą szybciej, taniej i bezpieczniej uczyć się nowej pracy.

Abstract for general public

Robots are increasingly trained in a "dream", a copy of the world predicted by a neural network called a world model, because real-world learning is slow, costly and sometimes unsafe. The network



learns from videos what usually happens after a robot moves, but its dream is never perfect. It may ignore a movement, let the gripper pass through an object or drift away from reality. A robot that does well in the dream often fails in reality, especially in new places. I want to find out which mistakes of the dream harm the robot and where inside its neural network they turn into wrong actions. I also want to check which few real trials best correct both the dream and the robot. I will study robot navigation and the grasping and moving of objects. The result should be methods that let robots learn new work faster, more cheaply and more safely.

11. Termin oddania do druku artykułu naukowego lub monografii / Date of submission for publication of at least one scientific article or monography

Termin wysłania do recenzji artykułu naukowego lub monografii naukowej zgodnie z wymaganiami art. 186 ust. 1 pkt 3 Ustawy stawianymi osobie ubiegającej się stopień doktora. Termin dotyczy oddania do druku co najmniej jednego artykułu naukowego opublikowanego w czasopiśmie naukowym lub w recenzowanych materiałach z konferencji międzynarodowej w odpowiedniej dyscyplinie naukowej, które w roku opublikowania artykułu w ostatecznej formie były ujęte w wykazie („lista ministerialna”) sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. b) lub 1 monografii naukowej wydanej przez wydawnictwo, które w roku opublikowania monografii w ostatecznej formie było ujęte w wykazie sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. a). / A date of submission for publication of at least one scientific article published in a scientific journal or in the peer-reviewed materials of an international conference in the relevant scientific discipline, which in the year of publication of the article in its final form were included in the list drawn up in accordance with the regulations issued under the Act (Art. 267 sec. 2 point 2(b) or 1 scientific monograph published by a publishing house, which in the year of publication of the monograph in its final form was included in the list drawn up in accordance with the regulations issued pursuant to the Act (Art. 267 sec. 2 point 2(a), referred to in Art. 186 sec. 1 point 3 of the Act.

September 2027 (planned submission to a venue worth 200 points on the ministerial list, such as the ICLR 2028 conference or the journal IEEE Robotics and Automation Letters).

12. Inne / Other comments

Dodatkowe wyjaśnienia, komentarze, podsumowanie związane z indywidualnym planem badawczym (maks. 1 strona, czcionka 11 pt., interlinia 1). / Additional explanations, comments, summary related to the individual research plan (at most 1 page, font 11, line spacing 1).

I used Claude (via Claude Code) for literature search, drafting and editing, and I reviewed all content and references myself.

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13. Określenie planowanej formy współpracy z promotorem / **Planned form of cooperation with the supervisor**

Krótki opis planowanej formy współpracy z promotorem, np. częstotliwość spotkań, forma spotkań i ich zakres, sposób komunikacji. / Brief description of the planned form of cooperation with the supervisor, e.g., frequency of meetings, form of meetings and their scope, method of communication.

My supervisor and I plan to meet weekly. At these meetings I will present my progress and recent results, and we will agree on the next steps and discuss our joint papers. I will also attend the weekly meetings of my supervisor's group of PhD students, where we discuss current results, problems and new work in the field. For daily contact we will use email and instant messaging. Code, trained models and the LaTeX sources of papers will be kept in shared Git repositories.

14. Opinia promotora pomocniczego / **Opinion of assistant supervisor**

Promotor pomocniczy (jeżeli jest wyznaczony) powinien zaopiniować IPB. / Assistant supervisor (if appointed) should provide an opinion of the IRP.

nie dotyczy / not applicable

.....



Podpis promotora pomocniczego /
Signature of assistant supervisor

15. Podpisy doktoranta i promotora/promotorów / Signatures of doctoral student and supervisor(s)

Wybierz datę/Select date

.....
Data / Date

.....
Podpis doktoranta / Signature of doctoral student

.....
Podpis promotora / Signature of supervisor

.....
Podpis drugiego promotora / Signature of second supervisor